

Assignment - 1

1.1) (a) Sample space of the two-coin-tossing experiment

$$\Omega = \{ HH, HT, TH, TT \}$$

(b) Event space is the set of all possible subsets of

$$\begin{aligned} \Omega \\ \{ \phi, \{HH\}, \{HT\}, \{TH\}, \{TT\}, \{HH, HT\}, \{HH, TH\}, \\ \{HH, TT\}, \{HT, TH\}, \{HT, TT\}, \{TH, TT\}, \{HH, HT, TH\}, \\ \{HH, HT, TT\}, \{HH, TH, TT\}, \{HT, TH, TT\}, \{HH, HT, TH, TT\} \end{aligned}$$

(c) (i) $P(\{HH\}) + P(\{HT\}) + P(\{TH\}) + P(\{TT\}) = 1$

Since all have
equal probabilities

$$p + p + p + p = 1$$

$$4p = 1$$

$$p = \frac{1}{4} = 0.25$$

(ii) Outcomes with atleast one head are
 $\{HH, HT, TH\}$

$$P(\text{atleast one head}) = \frac{3}{4} = 0.75$$

(iii) Outcomes with exactly one head are
 $\{HT, TH\}$

$$P(\text{exactly one head}) = \frac{2}{4} = \frac{1}{2} = 0.5$$

2.1) let A be the event when participant recognize 45 out of 50 words correctly.

$$f(k, n, p) = \frac{n!}{k! (n-k)!} p^k (1-p)^{n-k}$$

with $k=45$, $n=50$, $p=0.9$

$$\begin{aligned} P(A=45) &= \frac{50!}{45! \times 5!} \times (0.9)^{45} \times (0.1)^5 \\ &= 0.1849 \end{aligned}$$

2.2) (a) $f(k, \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$

$$\begin{aligned} \text{for } k=0, \lambda=10 \\ &= \frac{(10)^0 e^{-10}}{0!} \\ &= e^{-10} \\ &= 0.0000454 \end{aligned}$$

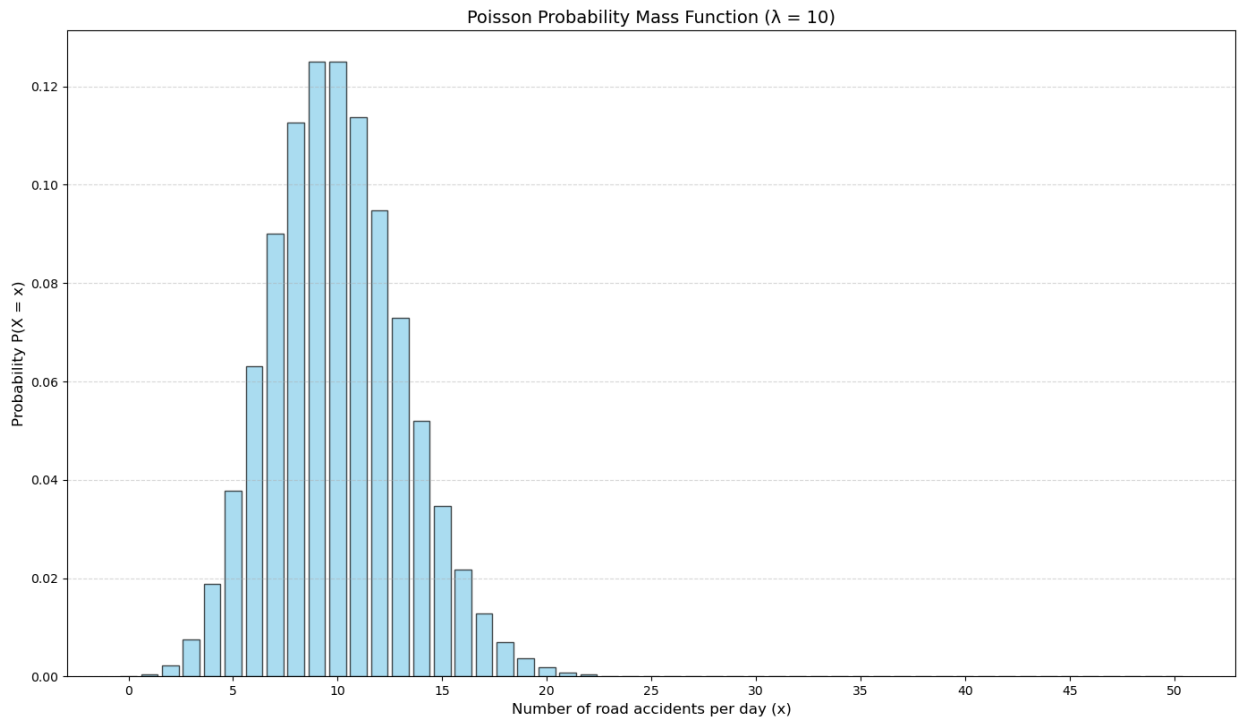
(b) $k > 7$, $k < 10$
that is $k=8$ or $k=9$

$$\begin{aligned} \text{for } k=8, \lambda=10 \\ &= \frac{(10)^8 e^{-10}}{8!} \approx 0.1126 \end{aligned}$$

$$\begin{aligned} \text{for } k=9, \lambda=10 \\ &= \frac{(10)^9 e^{-10}}{9!} \approx 0.1251 \end{aligned}$$

$$\begin{aligned} P(\text{more than 7 and less than 10 road accidents}) &= \\ &= 0.1126 + 0.1251 = 0.2377 \end{aligned}$$

(c)



3.1) (a) $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$\mu=1, \sigma=1$ $f(x=0) = \frac{1}{1\sqrt{2\pi}} e^{-\frac{(0-1)^2}{2(1)^2}}$

$$= \frac{1}{\sqrt{2\pi}} e^{-1/2}$$
$$= 0.24197$$

(b) $f(x=1) = \frac{1}{1\sqrt{2\pi}} e^{-\frac{(1-0)^2}{2(1)^2}}$

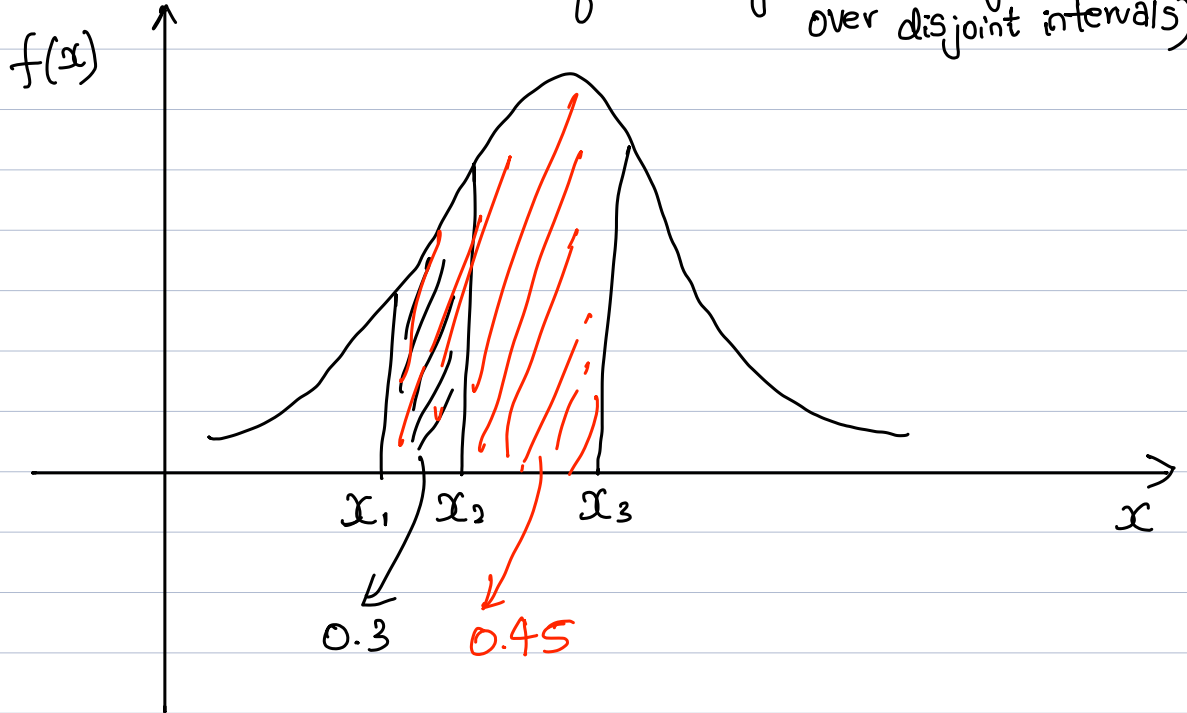
$$= \frac{1}{\sqrt{2\pi}} e^{-1/2}$$
$$= 0.24197$$

(c) Given,

$$P(x_1 \leq X \leq x_2) = 0.3$$

$$P(x_1 \leq X \leq x_3) = 0.45$$

If we visualise this in Pdf (Using additivity of probability over disjoint intervals)

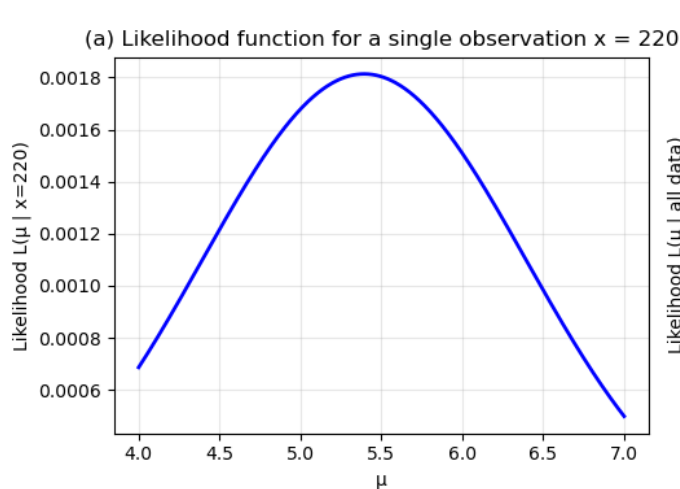


Assuming $x_3 > x_2$ since $0.45 > 0.3$

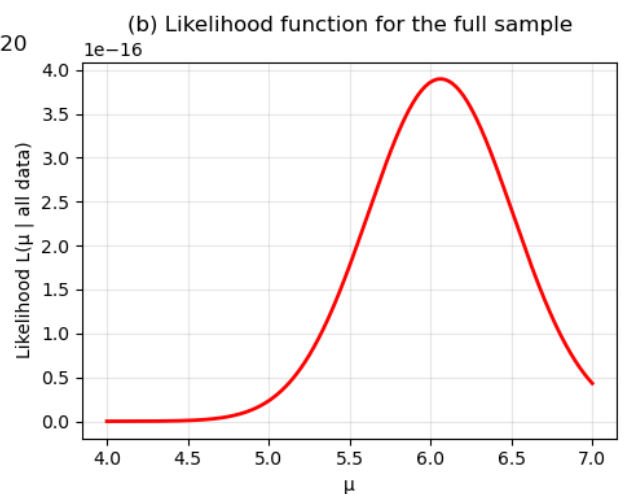
Clearly $P(x_2 \leq X \leq x_3) = 0.45 - 0.3$

$$P(x_2 \leq X \leq x_3) = P(x_1 \leq X \leq x_3) - P(x_1 \leq X \leq x_2) = 0.15$$

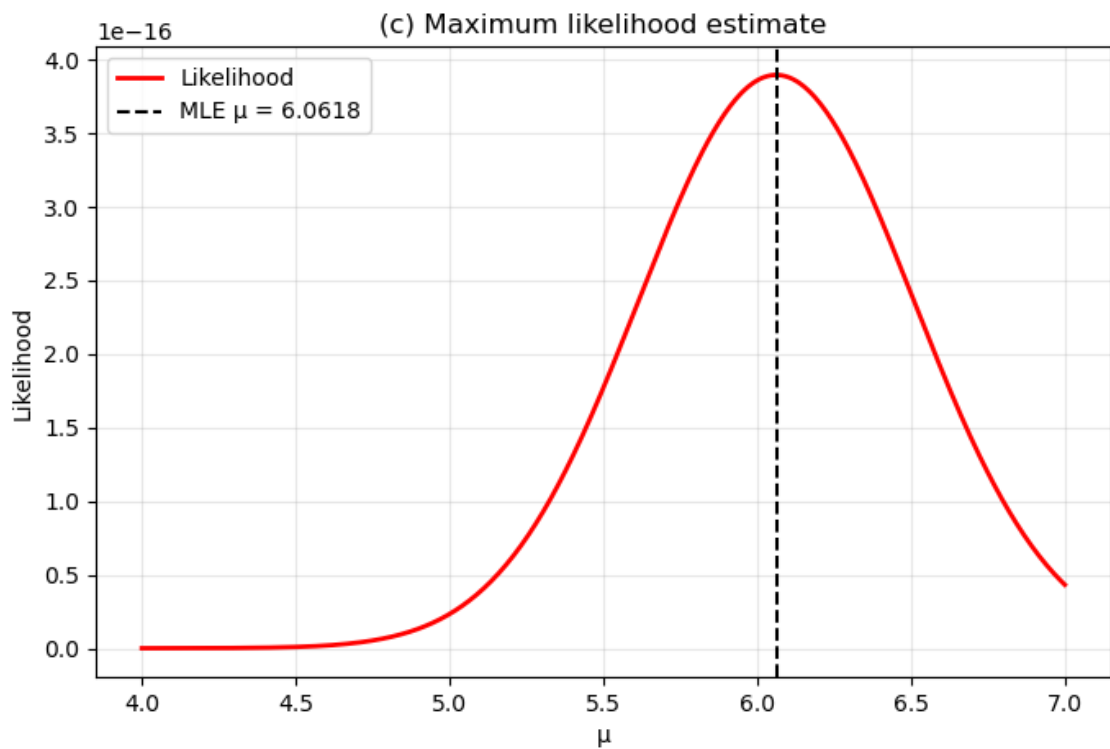
4.1)



(a)



(b)



for $\mu = 6.0618$, the likelihood of obtaining the observed sample 303, 443, 220, 560, 880 is maximum.

The code for generating the plots is given below

```

import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import poisson
from scipy.optimize import minimize_scalar

lambda_val = 10
x = np.arange(0, 51)
pmf = poisson.pmf(x, lambda_val)

plt.figure(figsize=(10, 6))
plt.bar(x, pmf, width=0.8, color='skyblue', edgecolor='black', alpha=0.7)
plt.xlabel('Number of road accidents per day (x)', fontsize=12)
plt.ylabel('Probability P(X = x)', fontsize=12)
plt.title(f'Poisson Probability Mass Function (lambda = {lambda_val})', fontsize=14)
plt.xticks(np.arange(0, 51, 5))
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()

x_single = 220
x_sample = np.array([303, 443, 220, 560, 880])

def likelihood_single(mu, x):
    return (1 / (x * np.sqrt(2 * np.pi))) * np.exp(-(np.log(x) - mu)**2 / 2)

def likelihood_joint(mu, x_array):
    n = len(x_array)
    log_lik = -n * np.log(np.sqrt(2 * np.pi)) - np.sum(np.log(x_array)) - 0.5 *
    np.sum((np.log(x_array) - mu)**2)
    return np.exp(log_lik)

mu_range = np.linspace(4, 7, 500)
L_single = [likelihood_single(mu, x_single) for mu in mu_range]

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.plot(mu_range, L_single, 'b-', linewidth=2)
plt.xlabel('mu')
plt.ylabel('Likelihood L(mu | x=220)')
plt.title('(a) Likelihood function for a single observation x = 220')
plt.grid(True, alpha=0.3)

L_joint = [likelihood_joint(mu, x_sample) for mu in mu_range]

plt.subplot(1, 2, 2)
plt.plot(mu_range, L_joint, 'r-', linewidth=2)
plt.xlabel('mu')
plt.ylabel('Likelihood L(mu | all data)')
plt.title('(b) Likelihood function for the full sample')
plt.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

log_means = np.mean(np.log(x_sample))

```

```

def neg_log_likelihood(mu, x_array):
    n = len(x_array)
    return n * np.log(np.sqrt(2 * np.pi)) + np.sum(np.log(x_array)) + 0.5 * np.
        sum((np.log(x_array) - mu)**2)

result = minimize_scalar(neg_log_likelihood, args=(x_sample,), bounds=(5, 7),
    method='bounded')

plt.figure(figsize=(8, 5))
plt.plot(mu_range, L_joint, 'r-', linewidth=2, label='Likelihood')
plt.axvline(x=log_means, color='black', linestyle='--', label=f'MLE mu = {
    log_means:.4f}')
plt.xlabel('mu')
plt.ylabel('Likelihood')
plt.title('(c) Maximum likelihood estimate')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()

```